

Membrane Characteristics

Let's now turn our attention to the membrane itself. We will explore the function of the membrane oxygenator, its limits, and how we can appraise and assess this function at the bedside. Understanding the unique characteristics of the membrane is going to be essential – the better we understand the strengths and limitations of the membrane, the better we will be able to leverage its capabilities when we are treating our patients.

Specifically, we are going to explore the concept of sweep gas flow and how adjustments to the rate of flow lead to changes in CO_2 . Recall the convention that we first introduced in the extracorporeal membrane oxygenation (ECMO) Fundamentals section on how to manipulate and adjust the ECMO circuit.

FiO_2 and ECMO blood flow mainly affect **oxygenation/oxygen delivery** while sweep gas flow affects **ventilation/ CO_2 removal**

We spent the last chapter exploring the effect of blood flow on oxygen delivery. Now we will answer the following question:

Why does manipulation of sweep gas flow mainly affect CO_2 clearance?

Answering this question will require an exploration of the membrane itself and the ways that sweep gas flow interfaces with the membrane.

WHAT IS THE BASIC FUNCTION OF THE MEMBRANE OXYGENATOR?

Let's return to our illustration of the membrane at the most basic level. You will recall that blood flows into the membrane, diffusing around plastic tubes that carry gas, which establishes a semi-diffusible barrier between a sweep gas phase and a blood phase (Fig. 11.1).

Much like the alveolar-blood interface, this blood/gas interface does not allow the diffusion of blood components but is diffusible enough to allow oxygen and CO_2 to travel down their concentration gradients. That means that if the FiO_2 is set to 100%, the PaO_2 of the sweep gas flow would be around 713 mmHg while the PaO_2 of the venous blood flowing into the oxygenator is much lower, say around 40 mmHg. Therefore, O_2 flows down its concentration gradient, from an area of high concentration (sweep gas) to an area of low concentration (blood). Even if the blood has a relatively high oxygen content, with a PaO_2 of 55 mmHg or even 60 mmHg, the gradient between sweep gas and blood is so high that oxygen will still diffuse, such that the PaO_2 of the blood will increase.

In a similar manner, CO_2 will move in the opposite direction out of the blood, since it exists at a higher concentration in the venous blood, say around 45 mmHg, versus 0 mmHg in the sweep gas.

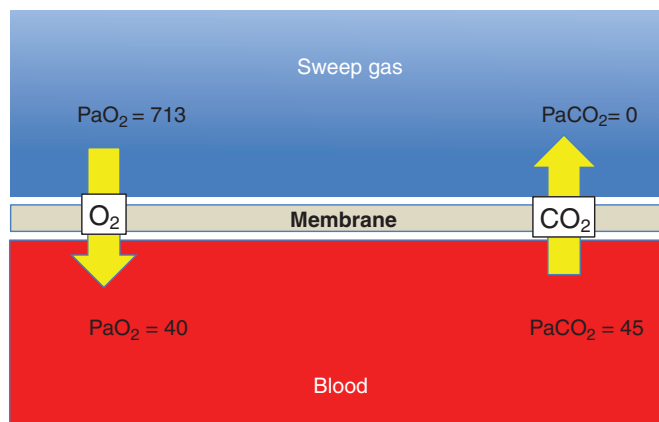


FIG. 11.1 The diffusion of O_2 and CO_2 across the membrane

With such large gradients existing between oxygen and CO_2 clearance, you can imagine that these gases are able to diffuse at high efficiencies.

IS ECMO MORE EFFICIENT AT OXYGENATION OR CO_2 CLEARANCE?

The answer here can be surprising. Many think oxygenation – it's called extracorporeal membrane *oxygenation* after all. Moreover, the relative gradient between PaO_2 in the sweep gas and in the blood is much greater. But the answer is CO_2 clearance. It's actually not even close – it is CO_2 clearance by a long shot. Why is that?

Efficiency of the Membrane: Effect of Relative Diffusibility

The main reason is diffusibility. CO_2 is more diffusible across the membrane. How much more diffusible? Six times more! This higher diffusibility allows for higher CO_2 clearance and overtakes the larger gradient that exists for oxygen between the sweep gas and blood.

Let's reflect on why this occurs. Imagine two lines at a toll booth to a bridge. One line is for cars with an automated pass. A sensor detects their pass and they can drive right through. The second line is for cars that have to pay cash. They have to stop, pay the operator, and receive change before proceeding on. If that first line is 6 times faster, it would not matter how many cars were in the second line; the concentration of cars from the first line on the bridge would be higher.

This is why sweep gas flow is so important to CO_2 clearance. The faster it flows, the more it clears CO_2 since CO_2 diffuses so much more rapidly.

Efficiency of CO_2 Clearance: Effect of Hemoglobin

The other reason that ECMO is so much more efficient at CO_2 clearance than oxygenation comes back to the difference between how oxygen is carried throughout the body and how CO_2 is carried throughout the body. You will no doubt recall from [Chapter 1](#) that the way that oxygen is carried throughout the body is through binding to hemoglobin. This is for good reason – if it wasn't bound to hemoglobin, it would just diffuse to tissues based on proximity and there would be no efficient distribution throughout the body. Like we have established before – SO_2 is more important to oxygen delivery than PaO_2 .

CO_2 is different. It does not need to be delivered and distributed evenly like oxygen does. Which means that CO_2 can simply remain dissolved in the blood, represented by $PaCO_2$.

Now that we have established these premises, let's examine the effect of the membrane on oxygenation and CO₂ clearance.

Let's start with oxygenation. Say you have a normal venous saturation of 70% of the blood entering into the oxygenator. This may correspond to a PaO₂ of 40 mmHg, which is much lower than the PaO₂ of 713 mmHg in the sweep gas, allowing for excellent diffusion of O₂ into the blood, so much so that the PaO₂ of blood leaving the oxygenator is 400 mmHg with a saturation of 100% (Fig. 11.2).

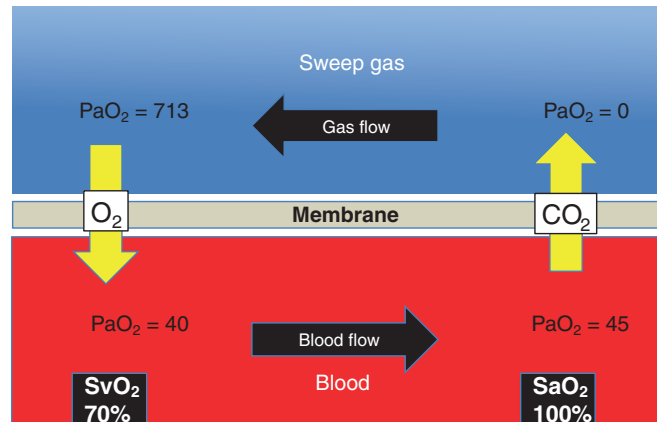


FIG. 11.2 The impact of hemoglobin saturation on membrane oxygen effectiveness

Now let's compare this to CO₂. In this case, the PaCO₂ of the blood entering into the oxygenator may be say 45 mmHg. This is significantly higher than the PaCO₂ of the sweep gas flow of 0 mmHg. Thus, the CO₂ molecules diffuse down their concentration gradient and the CO₂ is drawn down to <10 mmHg.

You can now see how CO₂ clearance can also be relatively more significant. Even though in this example, O₂ concentrations are increasing 10 fold and CO₂ concentrations are decreasing 10 fold, the parameter that matters to oxygen delivery of the body only increases by 30% (from 70% to 100%) (Fig. 11.3).

For oxygenation, you are limited by the saturation capacity of hemoglobin, while for CO₂ clearance, no such limit exists limiting PaCO₂.

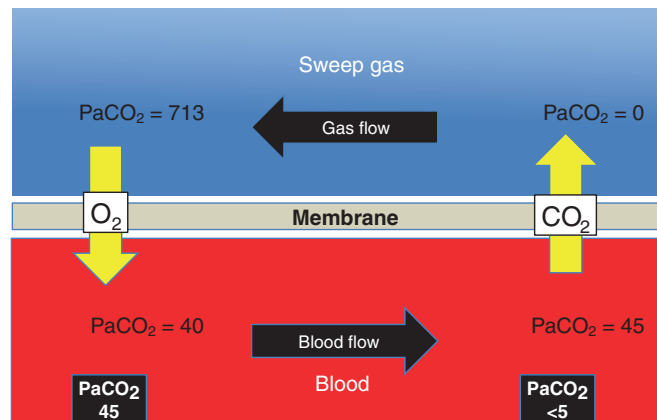


FIG. 11.3 The impact of dissolved CO₂ on membrane CO₂ clearance

LEVERAGING SWEEP GAS FLOW RATE

The sweep gas flow rate will allow for CO_2 to be swept away at a proportional rate. Therefore, the higher the rate of sweep gas flow, the higher the CO_2 clearance.

This works out in a favorable manner when it comes to how sweep is titrated. It means that the only limit to how high this gas rate can be increased is the capacity of the oxygenator/blender, which is usually 10–15 L/min.

Compare this to the limits of blood flow that we discussed in the prior chapter – everything from anatomic limitations to the limitations of the circuit/oxygenator to the preload dependence/afterload sensitivity of the pump to hemolysis to recirculation can limit blood flow. For sweep, it is a matter of turning a dial and increasing gas flow.

You can see that increasing sweep gas flow rates by 2–3 $\times$ can easily be done, which cannot always be said about ECMO blood flow.

Does This Mean That Blood Flow Through the Membrane Has No Effect on CO_2 Clearance?

Blood flow actually can have a significant effect on CO_2 clearance. Higher blood flows translate into higher CO_2 clearance to a certain point, in a similar manner to how higher blood flow translates into higher oxygen delivery (Fig. 11.4).

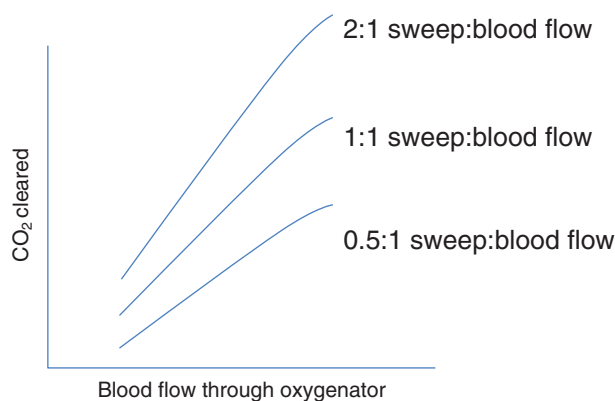


FIG. 11.4 CO_2 clearance as a function of sweep gas flow and blood flow

However, as compared to the oxygenation curve, this curve becomes steeper and accounts for more CO_2 clearance as the sweep gas flow increases compared to blood flow increases.

What does this mean clinically? Should we be targeting blood flow when we are attempting to optimize CO_2 clearance? Since sweep gas flow changes can be made so readily, my approach is to arrive at the ideal blood flow that is needed to meet oxygen delivery requirements and then adjust sweep gas flow accordingly to dial in the CO_2 clearance needs.

LIMITATIONS OF THE MEMBRANE OXYGENATOR

All of this assumes a normally functioning membrane oxygenator. An oxygenator can perform suboptimally for a number of reasons, from how blood is flowing to the function of its fibers. We will now dive deeper into the issues with membrane oxygenator function and review some of the patterns for assessing how the membrane loses effectiveness.

Effect of Thrombus Formation on the Function of Oxygenator

As time on ECMO increases, the performance and function of the oxygenator tends to decline. The primary reason for this decline involves the activation of a host of inflammatory and coagulation cascades as blood moves through the oxygenator. These cascades can lead to microscopic and larger clot formation on the fibers of the oxygenator that can literally disrupt the flow of gases across it (Figs. 11.5 and 11.6).

However, this process can often be quite unpredictable; some oxygenators can perform optimally for weeks to months, while others can last only a few days before showing signs of poor performance. This partially occurs due to variations in anticoagulation status, blood flow rates, and occurrence of disruptions in flow. However, there are likely a number of host factors that are not yet fully understood that also contribute to the upregulation of clot formation. We will discuss the regulation of the coagulation cascade in greater detail in Chapter 20.

Decreasing performance of the oxygenator due to clot formation can affect membrane function, alter the flow of blood, or cause damage to blood.

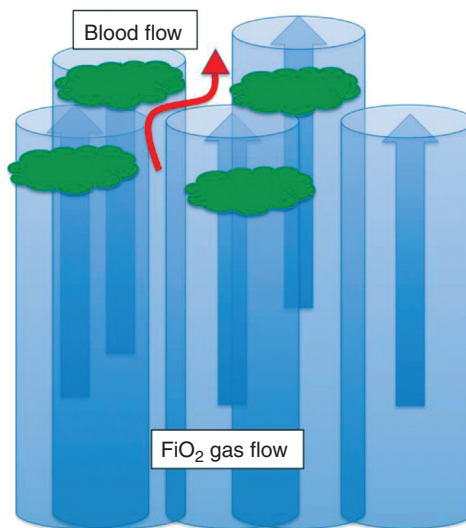


FIG. 11.5 Oxygenator thrombus formation

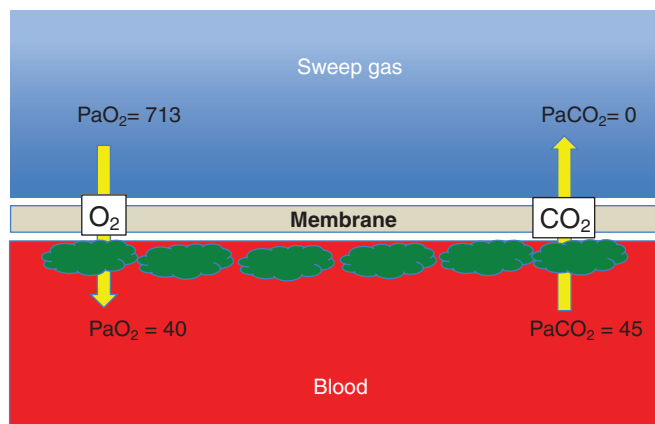


FIG. 11.6 Membrane oxygenator thrombus formation

EVALUATION OF OXYGENATOR FUNCTION

Evaluating the function of the oxygenator requires an assessment of the blood going in and out of the oxygenator. Often this comes in the form of assessing blood gases from three sources: the patient, pre-oxygenator, and post-oxygenator, as illustrated in Fig. 11.7.

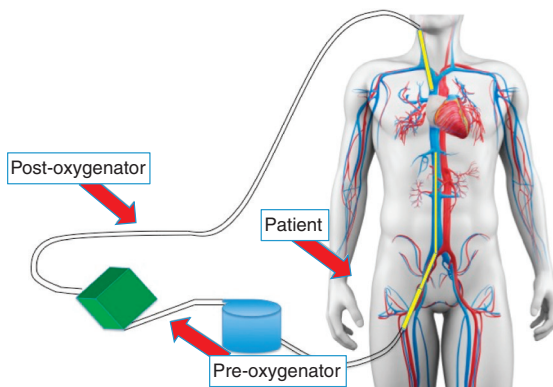


FIG. 11.7 Assessing oxygenator function with blood gas measurements: patient, pre-oxygenator, and post-oxygenator. (Modified from [SciePro/Shutterstock.com](https://www.shutterstock.com/user/sciepro).)

The reason for checking these gases is to give an adequate sense of the performance of the oxygenator and the clinical situation of what is going on with the patient.

1. Patient ABG: used to assess what is going on with the patient. Any changes to support, sweep gas flow, or blood flow are ultimately made to alter the parameters of this gas.
2. Post-oxygenator ABG: gives a sense of the performance of the oxygenator. The most important parameter is the PaO_2 , but the PaCO_2 is also taken into consideration.
3. Pre-oxygenator: used for calibrating SvO_2 monitors, assessing for recirculation, approximating VO_2 (with limitation), and contextualizing CO_2 clearance.

Checking all three sources (known as “triple gases”) is not necessary with every gas, but rather should be done if the desired effect is to examine both what is happening clinically with the patient and how the oxygenator is performing.

EVALUATION OF CO_2 CLEARANCE

CO_2 clearance (the difference in PaCO_2 pre-oxygenator and post-oxygenator) can give a sense of how the oxygenator is performing. Factors affecting CO_2 clearance include the PaCO_2 of the blood entering the oxygenator, the performance of the membrane, and the sweep gas flow rate. Thus, if PaCO_2 is relatively low entering the oxygenator, there may not be as much clearance of CO_2 since the PaCO_2 gradient between the blood and sweep gas is lower. Accordingly, if sweep gas flow is low, say because you are weaning down, the CO_2 clearance may be low. This is not due necessarily to a failing oxygenator, but rather due to a low rate of clearance due to the sweep gas flow rate.

As oxygenators start to have progressive thrombotic burden and a worsening performance, this can lead to a decreased diffusion of gases, which can manifest as a drop in CO_2 clearance.

Importantly, this decrease in CO_2 clearance may manifest before a drop in oxygenation. The high diffusibility of CO_2 under normal circumstances necessitates that a drop in the performance of the membrane may have a larger observable effect in CO_2 clearance. I start to consider oxygenator failure when the CO_2 clearance between pre-oxygenator and post-oxygenator is less than 10 mmHg despite being on a moderate amount of sweep (say 4–5 L/min).

EVALUATION OF OXYGENATION

Normally, PaO_2 is of less concern when we are evaluating blood gases, because saturation is the marker of adequate oxygen delivery. However, remember that the performance of the membrane oxygenator is dictated by how well gases can diffuse. Therefore, when evaluating the post-oxygenator gas, the PaO_2 is paramount, as it alone dictates how well the oxygenator is performing when it comes to allowing the diffusion of oxygen.

You should anticipate a high post-oxygenator PaO_2 if on 100% FiO_2 due to the high gradient that exists from the sweep gas (PaO_2 713 mmHg) to the blood (PaO_2 ~40 mmHg) (Fig. 11.8).

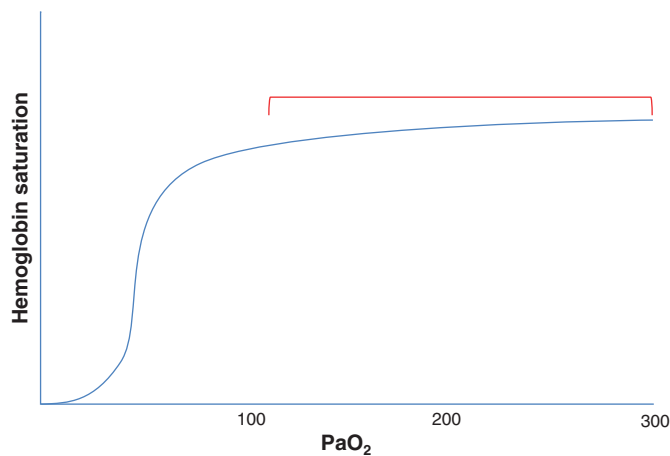


FIG. 11.8 Minimal impact of high PaO_2 on hemoglobin saturation

Additionally, as you may recall from our hemoglobin dissociation curve there is a flat portion of the curve where PaO_2 increases but has a negligible effect on saturation. Although this PaO_2 does not improve the delivery of oxygen significantly, monitoring for changes in the PaO_2 , say from 450 to 200 over the course of a couple of days, may be a significant indicator that the oxygenator function is worsening, even if the saturation of the blood leaving the oxygenator remains at 100%.

EVALUATION OF MEMBRANE FUNCTION: DAMAGE TO BLOOD CELLS/CONSUMPTION

An important part of the evaluation of the function of the oxygenator is the effect on the blood itself. As clot burden increases, these clots can exert a host of adverse effects on the blood that is circulating through the membrane.

One such effect is damage to the cells itself, primarily red blood cells. Close evaluation for markers of hemolysis (jaundice, elevated plasma-free hemoglobin, elevated lactate dehydrogenase (LDH), elevated bilirubin, and dark urine) can help to identify worsening oxygenator function, particularly if revolutions per minute (RPMs) to the pump have remained relatively stable.

Thrombus and clot formation can have the additional adverse effect of upregulating more clot formation, therefore exerting a consumptive effect of the membrane. In this case, there may be a consumptive coagulopathy with a drop in fibrinogen, coagulation parameters, and platelets similar to disseminated intravascular coagulation (DIC) but occurring within the oxygenator rather than intravascularly.

Evidence of any of these phenomena point toward a worsening oxygenator function, even in the presence of adequate gas exchange, and is thus often part of the daily evaluation of the circuit.

EVALUATION OF MEMBRANE FUNCTION: EFFECT ON MEMBRANE BLOOD FLOW AND PRESSURE

The third manifestation of a failing membrane is a change in pressure and blood flow. This is largely due to thrombus formation on the oxygenator, which impedes flow by adding afterload to the pump. The added afterload decreases the pressure differential that the pump is able to generate, and eventually leads to a drop in flows. This drop can be gradual or precipitous. Usually, alteration in pressures is a late manifestation of oxygenator failure, but it can happen right after cannulation, likely due to the effect of an existing thrombus that gets sucked into the oxygenator by the drainage cannula.

If pressures are being measured/transduced, then transmembrane pressures measured pre-oxygenator and post-oxygenator can help to identify oxygenator thrombus and worsening function.

To review our process for evaluating membrane pressures, we can measure circuit pre-oxygenator, post-oxygenator, and pre-pump (if you have an integrated sensor). The highest pressure of the circuit will be the pressure directly after the pump (pre-oxygenator), as this is the highest pressure that will be generated in the circuit. After this, the pressure drops across the membrane, to a degree that is equal to the transmembrane pressure or delta P. The pressure post-oxygenator then drops gradually until it equals the mean arterial pressure (MAP) or central venous pressure (CVP), depending on if you are on VA or VV ECMO (Fig. 11.9).

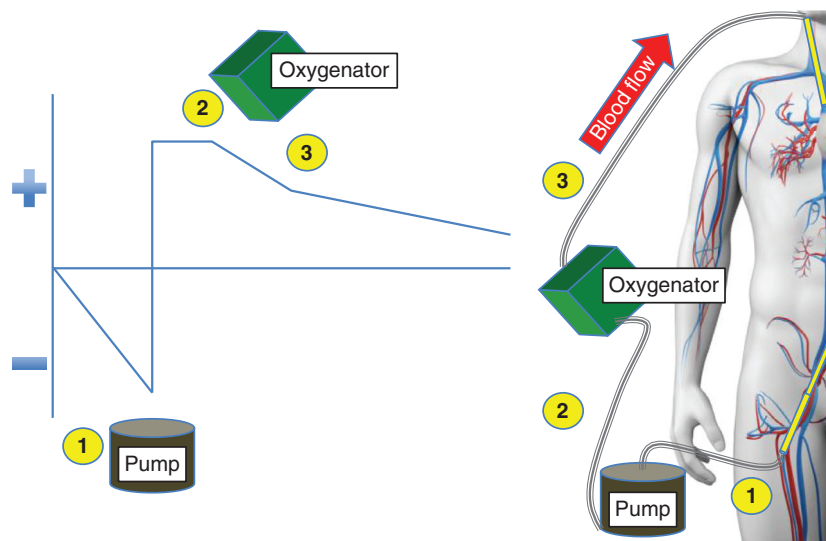


FIG. 11.9 Pressures throughout the ECMO circuit. (Modified from [SciePro/Shutterstock.com](https://www.shutterstock.com/user/sciepro).)

We can now contrast this to the below situation, where there is a thrombus formation on the oxygenator. In this case, the pressure needed to maintain adequate flow rises precipitously, otherwise you would experience a drop in flows. You will note that the pressure at position 3 (post-oxygenator) remains the same; the escalation in delta P results from an increase in pre-oxygenator pressure due to the obstruction across the oxygenator (Fig. 11.10).

Escalation in membrane pressures, especially rapid escalation over the course of minutes to hours, should be intervened on rapidly as this can be a harbinger of impending circuit failure.

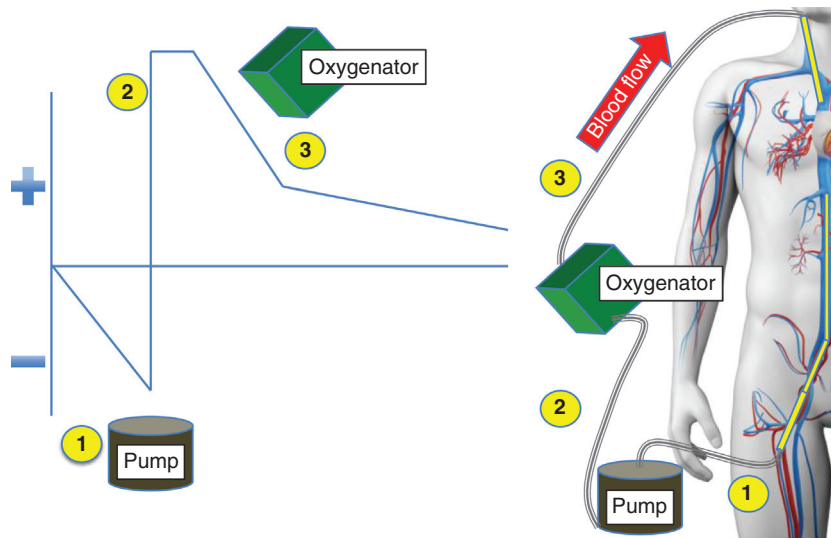


FIG. 11.10 Impact of oxygenator thrombus with elevated post-pump/pre-oxygenator pressure. (Modified from [SciePro/Shutterstock.com](#).)

FAILURE OF MEMBRANE FUNCTION DUE TO RECIRCULATION

Let's return to the concept of recirculation and explore its implications for affecting membrane function. As you'll remember, recirculation is a circumstance unique to VV ECMO, which involves blood that is being returned from the circuit being sucked back into the circuit before it has the opportunity to be circulated to the rest of the body (Fig. 11.11).

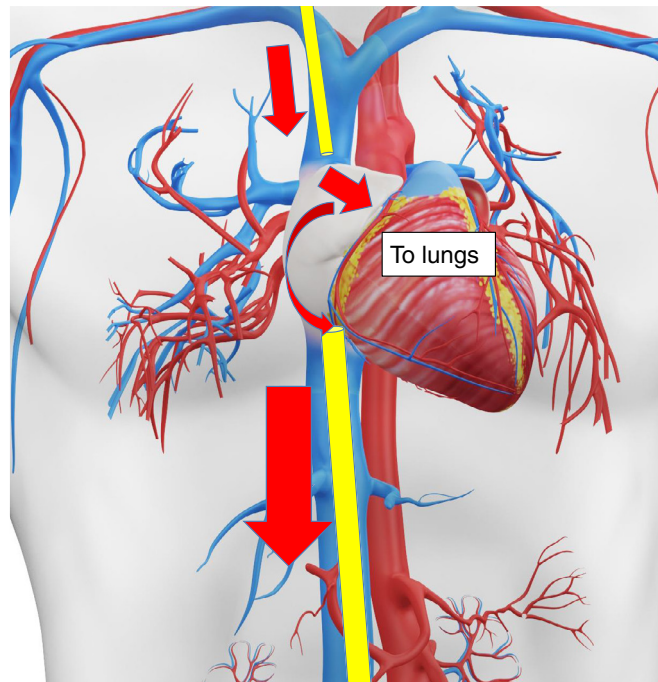


FIG. 11.11 Recirculation of blood. (Modified from [SciePro/Shutterstock.com](#).)

You will recall that the first adverse effect of recirculation is a decrease in effective blood flow, since any volume of blood that is recirculated is simply going back to the circuit and not being distributed to the body.

The causes of recirculation are as follows:

1. Excessive blood flow rates: the high negative pressures required to maintain these flows exert more pull on returning blood.
2. Cannula malposition: lack of adequate space between the return and the drainage cannula can allow for returned blood to get sucked back into the drainage cannula.
3. High negative intrathoracic pressures: coughing or deep breathing, with strong exertion of the diaphragm, can lead to a negative pressure causing recirculation either continuously or rhythmically, with an increase in recirculation with each breath.
4. Right ventricular failure: due to returned blood not being adequately circulated into the pulmonary circulation, backing up, and then being sucked into the drainage cannula

How Does Recirculation Adversely Affect Membrane Function?

Recirculation also affects membrane function, rendering it less effective. Let's consider an example to illustrate.

Your patient is on VV ECMO. You are flowing 3 L, with a venous saturation of 60% and a patient saturation of 90%. Hoping to improve the saturation, a well-meaning member of the staff increases the blood flow to 5 L. The saturation decreased to 85% and the venous saturation increased to 80%.

What happened here? Why did the saturation go *down*?

This example illustrates the two adverse effects of recirculation. The first is that the effective flow went down, as the higher blood flow pulled more blood into the drainage cannula, and less was able to flow to the patient.

The second is how recirculation decreases the effectiveness of the membrane. Prior to going up on flow, the membrane was able to raise the saturation of the patient 30%, from 60% to 90%. But raising the saturation of the drained blood decreases the amount that the membrane can oxygenate, which is now only 5%.

Recirculation adversely affects the patient by decreasing the effective blood flow **and** by decreasing the effective function of the membrane.

PUTTING IT TOGETHER

Understanding the membrane function is an essential component of understanding the physiology of ECMO support. Like most aspects of ECMO, membrane function requires serial attention to trends. Evidence of membrane failure can be more readily recognized when placed into the context of what is happening with the patient on a daily basis.

SUGGESTED READING

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